Experiment Name: 8-bit Subtraction using Assembly Language

1. Objective

The objective of this experiment is to perform 8-bit subtraction using Assembly Language in the 8086 microprocessor. This experiment helps to understand how arithmetic subtraction operations are performed using registers and instructions.

2. Apparatus / Tools

- Personal Computer (PC)
- EMU8086 Emulator Software

3. Working Procedure

Steps:

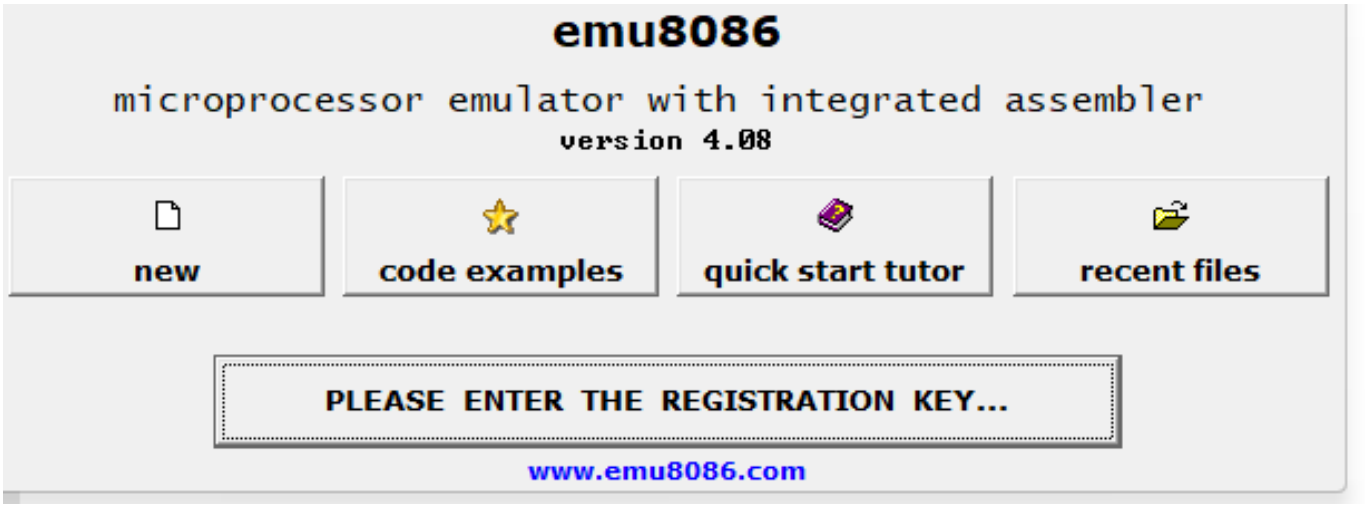
1. The first number is loaded into the AL register.
2. The second number is loaded into the BL register.
3. The SUB instruction is used to perform subtraction.
4. The result is stored in the AL register.

4. Program:

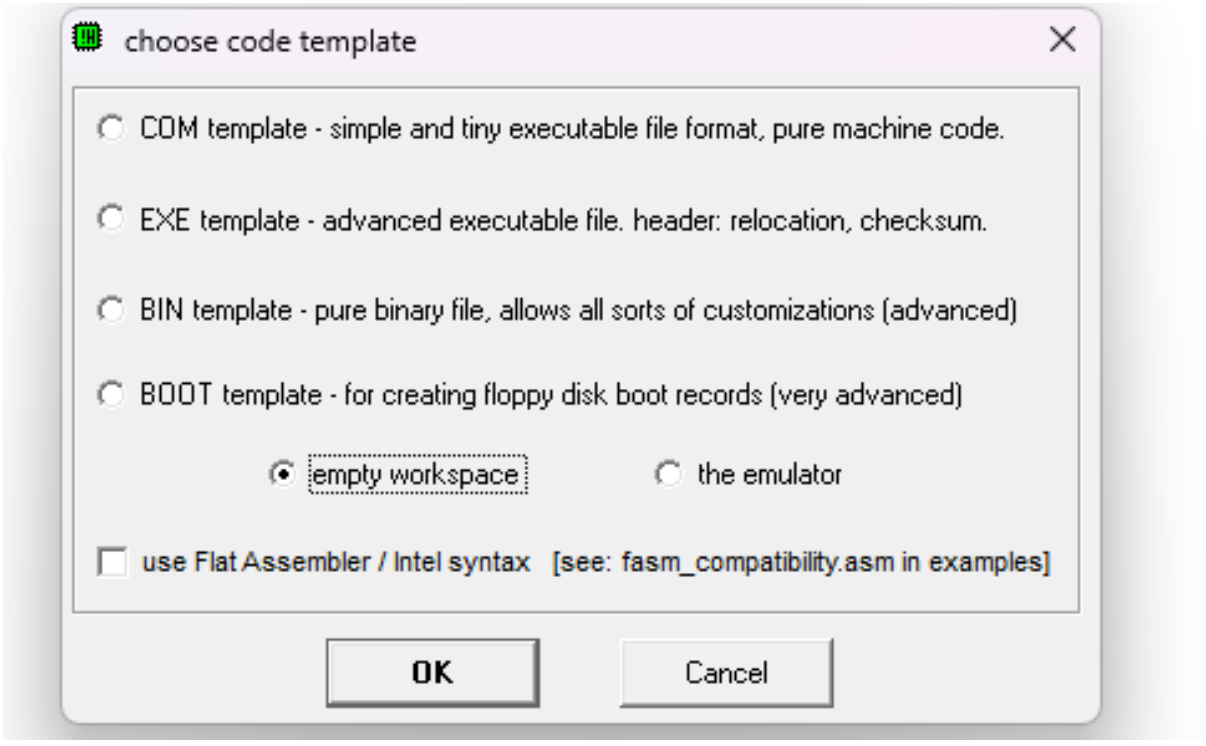
```
.MODEL SMALL
.STACK 100H
.DATA
.CODE
MAIN PROC
 MOV AL,20H
 MOV BL,10H
 SUB AL,BL
 INT 21H
 MAIN ENDP
END MAIN
```

5.Operation Step

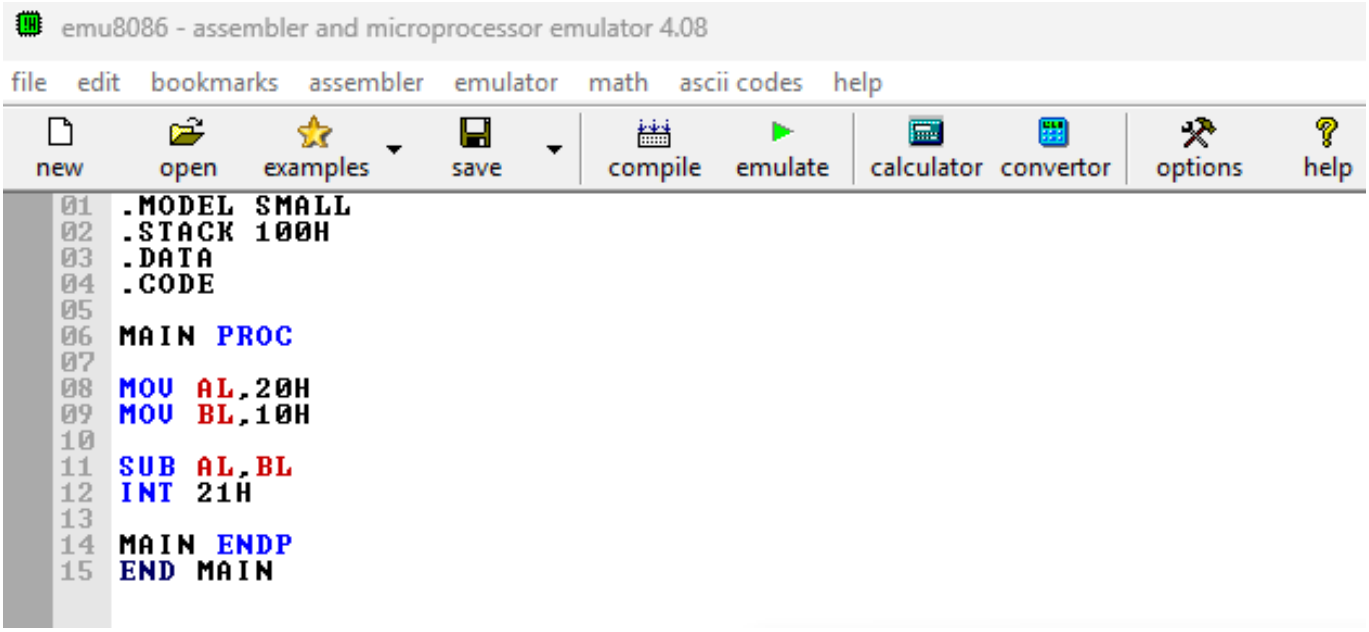
Step 1: Open EMU8086 and click on 'New' to start a project.



Step 2: Select “Empty Workspace”
After clicking on “New”, a new window will appear. From there, select “Empty Workspace” and click OK to proceed.

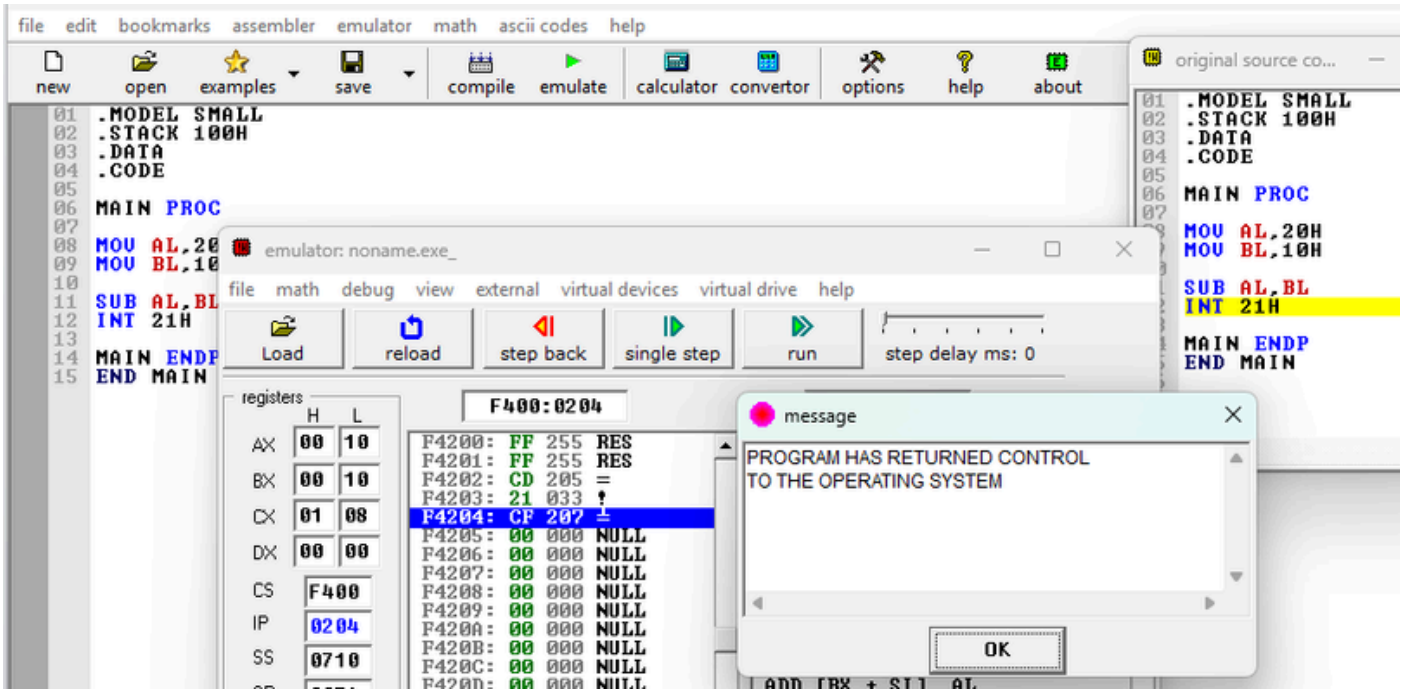


Step 3: Write the Program Code
After opening the workspace, write the assembly language program in the editor window.



5.Operation Result

Step 1: Click on the Run/Execute option to run the program.
The result of the subtraction will be shown in the AL register.



6.Comment

The program executed successfully, and the 8-bit subtraction was correctly performed. The result (10H) was stored in the AL register.

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